

HLA pangenome: a three-day research sprint

High-level plan, v1.1.0

Robert Hoehndorf, Dawn, and Hassan

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1. The question and the intended outcome

Does a diverse HLA graph improve the recovery and typing of HLA alleles from short reads, and how much reference diversity is needed? We will compare the same 1000 Genomes samples against hg38 and our graph, using experimental HLA types to assess accuracy. The graph starts from 610 haplotypes; adding Korea and CRC brings the planned panel above 700. The panel emphasises Asian diversity and includes non-Asian HPRC donors.

Three-day outcome: a working comparison on a independently typed sample set, a first comparison matrix, error and cost analyses, and initial haplotype and bundle results. Completion of each analysis depends on data readiness and measured cluster throughput. The schedule assumes concurrent work by Robert, Dawn, Hassan, and agents, with access to the cluster and existing data.

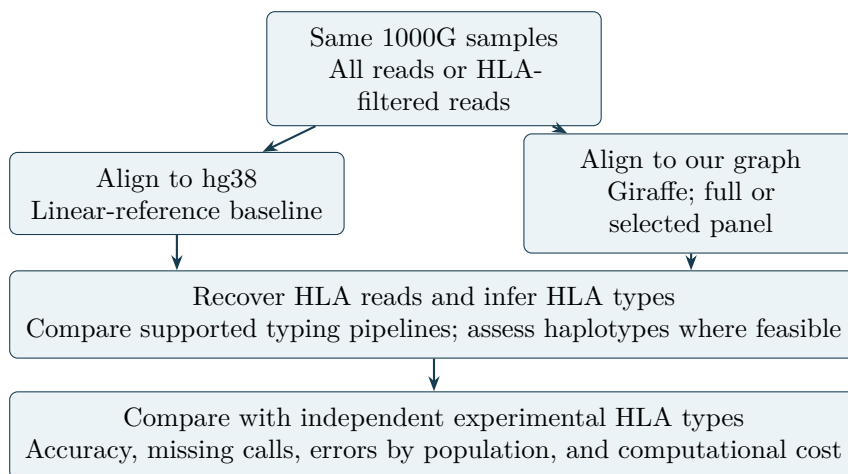


Figure 1: Core experiment. Graph construction and truth-data preparation proceed in parallel. Experimental labels are used only for evaluation.

2. Keep the first comparison small

Build and check. Consolidate the assembly panel, build or reuse the HLA graph, and select an initial pilot of approximately 12–20 available, independently typed individuals spanning represented populations. Exclude their own assemblies and close relatives from the graph. Use the pilot to fix the pipeline and then launch a larger held-out cohort as soon as cluster throughput permits.

Compare and explain. Begin with hg38 versus the full eligible graph and a shared read set. Then test whether HLA filtering preserves the calls while reducing cost. Start two compatible HLA typers and add the third in parallel: HLA*LA, T1K, and SpecHLA. Compare calls at the resolution supported by the experimental assay, count missing calls, and inspect discordant cases.

3. Three-day timeline: parallel work, early compute

Plan for agent-assisted implementation at approximately one conventional week per day, as a working assumption. The critical path is data access, graph construction/indexing, and cluster runs. Owners below are proposed.

Day	Concurrent workstreams	Checkpoint and output
1	Dawn + agents: panel QC, graph build/indexes, subset jobs. Hassan + agents: experimental truth, read extraction, three typer wrappers. Robert + agents: scoring, inference prototype, experiment manifest.	First hours: verify data, exclusions, and one end-to-end sample. Launch pilot and expensive graph jobs immediately; submit bulk runs once the pilot passes. Leave the cluster processing overnight.
2	Dawn: mapping and subset runs. Hassan: typing and paired evaluation. Robert: error review and inference validation. Agents: expand job arrays, test the prototype, and analyse existing bundles while compute runs.	First accuracy/cost comparison; full matrix submitted where resources allow. Freeze settings before held-out evaluation. Test matched long reads if already available. Schedule failed-job retries overnight.
3	All three owners + agents: complete runs, inspect discordant calls, analyse population effects and subset sizes, assess haplotype and bundle results, and write the report.	Versioned graph, workflows, comparison matrix, uncertainty estimates, error examples, and exploratory figures. Identify unfinished compute and the specific evidence needed next.

Execution rule. Agents prepare code, manifests, tests, and reports concurrently; cluster arrays handle heavy computation. Give each output directory one owner. Share read-only indexes and cache extraction. Measure a representative job on day 1, then set concurrency and cohort size from actual memory, runtime, storage, and queue capacity.

4. Target comparison matrix

Axis	Planned conditions
Reference	hg38; full eligible graph; subsets of 4, 8, 16, 32 individuals.
Read input	All reads; HLA-filtered reads plus every unmapped read and mates.
HLA typer	HLA*LA, T1K, SpecHLA, using each tool's supported inputs.
Selection	Read-guided donor subsets and random controls. Test blockwise haplotype selection separately if indexes are ready [1].

Prioritise full graph versus hg38, then filtering, then subset size and selection. Whole-individual subsets retain both available haplotypes; the cited method selects haplotypes locally in blocks. Avoid duplicate runs when a typer receives identical reads and realigns them internally.

Fallbacks follow compute. If Korea/CRC indexing is delayed, start with the available donor-excluded panel and add the expanded graph when ready. If all-read mapping is expensive, complete it on a smaller paired cohort while filtered runs continue. Report the actual panel, samples, and completed cells for every comparison.

5. Assessment: realistic and worth pursuing?

My recommendation is to pursue it. The existing assembly panel and analysis work provide a strong starting point. With agents working concurrently, three days is a credible target for a substantial first benchmark and exploratory results. Completion is conditional on data readiness and measured compute time. Assessing rare alleles and population-specific effects depends on the number of independent, experimentally typed samples available.

The strongest scientific question is the value of added diversity. HLA graphs already exist [2]. This study becomes compelling if the expanded panel improves difficult calls, reduces errors in underrepresented populations, reconstructs additional sequence, or lowers cost. Compare panels with and without Korea/CRC while controlling panel size to identify what these additions contribute. A larger graph alone provides a resource; measured benefit provides the stronger research argument.

The hardest step is connecting graph alignment to HLA inference. Many typers realign recruited reads to their own references. Their comparison therefore tests read recruitment. A direct graph-inference experiment requires an inference stage that uses graph alignments. Assign an agent workstream to prototype it immediately, validate it on known sequences, and report its results separately from established typers. Keep both routes so the benchmark progresses while inference development proceeds.

The decisive controls are independent truth and donor exclusion. Remove evaluation donors and close relatives from the graph. Match output resolution to the experimental assay [3]; retain ambiguous and missing calls. Review the paired errors before interpreting an accuracy difference. Use the pilot for debugging and the held-out samples for evaluation.

6. Extensions to start in parallel

Haplotypes and long reads. Infer supported sequences within individual HLA genes first. Short reads provide local phase information; extended MHC phasing requires longer links or explicit reference-panel assumptions. I expect accurate long reads to help most with full-gene reconstruction and phasing. Test available matched individuals with SpecImmune and HLAMiner, controlling coverage and scoring resolution. Mark absent datasets as dependencies.

Bundle classification. Start from the existing bundle analyses on day 1 or 2. Cluster shared sequence segments without using HLA labels, then measure recovery of known types on held-out donors. This is a worthwhile exploratory result. The more ambitious question is whether clusters capture peptide-binding similarity or structural features beyond conventional HLA groups. Compare binding-site sequences first; use independent binding evidence as it becomes available. Population frequencies give context; phenotype associations need individual-level outcomes and ancestry adjustment.

Day 3 decision. Expand the most reproducible accuracy, cost, or biological finding. If typing performance is equivalent, examine efficiency and sequence coverage. If the graph-inference prototype is promising, define its next validation set. Publish a clear account of completed comparisons and outstanding compute, together with a prioritised next experiment.

References

- [1] Sirén J et al. Personalized pangenome references (2024). Nature Methods, doi:10.1038/s41592-024-02407-2.
- [2] Diltthey Lab. HLA*LA: graph-based HLA typing and supported inputs.
- [3] Gourraud P-A et al. HLA diversity in the 1000 Genomes dataset (2014). PLOS ONE, doi:10.1371/journal.pone.0097282.